

Embedding Agentic Traces

A trace representation should be faithful to content and invariant to authorship

Helivan Research — working summary, August 2026

The problem

Modern evaluation produces *trajectories*: an agent works on a software issue for tens of minutes and leaves a log of 10^4 – 10^5 tokens. Public archives already hold them at scale (SWE-bench Verified: ~ 119 systems $\times$ 500 instances, in dozens of incompatible formats), yet each run is reduced to one pass/fail bit. The obvious move—feed traces to an off-the-shelf embedder—measures the wrong thing: raw trace embeddings are *fingerprint readers*. Nearest neighbors are same-system 68–83%, same-scaffold at $5\times$ chance, and the exact agent configuration is retrievable at 0.82 vs. 0.43 chance even among same-model, same-harness siblings—while outcome coherence sits at chance (AUC 0.52). Geometric fixes fail: per-instance centering alone *amplifies* provenance dominance ($0.68 \rightarrow 0.83$), and a dozen unsupervised variants of the naive representation are performance-identical—an information ceiling.

qubric: similarity as an input

For each instance, an LLM reads the *public problem statement* (no labels) and writes a query-specific rubric: what understanding, localization, reproduction, editing, verification, and final state concretely mean for *this* issue. A judge LLM reads each trace through the rubric and extracts a short factual description per section; sections are embedded separately, centered against the cross-system median per instance (consensus centering), and concatenated. The rubric is a natural-language definition of the equivalence relation the space should respect; invariance comes from *re-observation through a fingerprint-poor channel*, not surgery on a fingerprint-rich one.

Evidence: the right sensitivity profile

Across **seven embedders** (three families, two orders of magnitude of size) qubric produces the same profile shift: task fidelity retained (~ 0.95), behavior retained or improved, and every authorship axis collapses to $\approx$ chance: provenance $0.83 \rightarrow 0.09$, scaffold $0.31 \rightarrow 0.08$, model lineage $0.20 \rightarrow 0.12$ (Fig. 1). Pillar independence is verified by double dissociation across six representations. The instance-conditioning is the active ingredient—the construction ranking is strict under both readouts (qubric > verdict questions > generic rubric > free-form summary > raw) and reproduces under a local embedder. No single rubric section carries the effect (best single section 0.091 vs. 0.078 for the union).

Judge robustness: quality degrades gracefully with

judge capability; an open frontier judge (DeepSeek-v3.1: 0.083/0.079) matches the closed reference (gpt-5.4-mini: 0.078/0.078) within panel resolution, and construction beats capability—a weak judge *with* a qubric outperforms a strong judge without one. **Reliability is measured:** two judges agree on a single trace only 12–28% of the time, but aggregating ~ 20 instances drives system-level agreement to 0.82–0.95. Per-trace noisy, per-agent reliable—and per-agent is what evaluation uses.

Payoff: query-efficient benchmarking

Qubric representations enter the DKPS/PKPS perspective space; a new agent’s full-benchmark resolve rate is predicted from a few probe traces against references’ cached coverage, blended with the sample score. Evaluation is leave-one-LLM-out throughout (naive CV inflates accuracy $\sim 25\%$ via model-family leakage).

probes m	sample score	qubric pipeline	+selection
1	0.44	0.095	—
5	0.16	0.074	0.057
20	0.06	0.048–0.053	0.053

Table 1: MAE predicting true resolve rates (107 systems). One probe trace ≈ 8 –10 scored runs.

One trace carries far more than one bit: the strongest correctness-only competitor (CAPA-style error overlap) needs 15–20 *scored* probes to match one *trace* (0.37 MAE at $m=1$), and trails the ensemble at every budget. The reference pool, not probe count, is the binding resource: error falls $0.145 \rightarrow 0.078$ as the pool grows $5 \rightarrow 96$, still declining (Fig. 2). Every layer is ablated; CV-inside-selection and PCA-composed-with-selection are documented overfitting traps.

Status & caveats

All qubric numbers: 20-instance panel, one benchmark; differences < 0.01 near panel resolution. The 150-instance scale-up is specified and unblocked (open-judge stack validated via OpenRouter; $\sim \$40$ – 85). Terminal-Bench (142 systems, 89 tasks, ≥ 5 trials/task, public trajectories) is scoped and deferred. Full log: FINDINGS.md (F1–F24) on the **agentic-traces** branch.

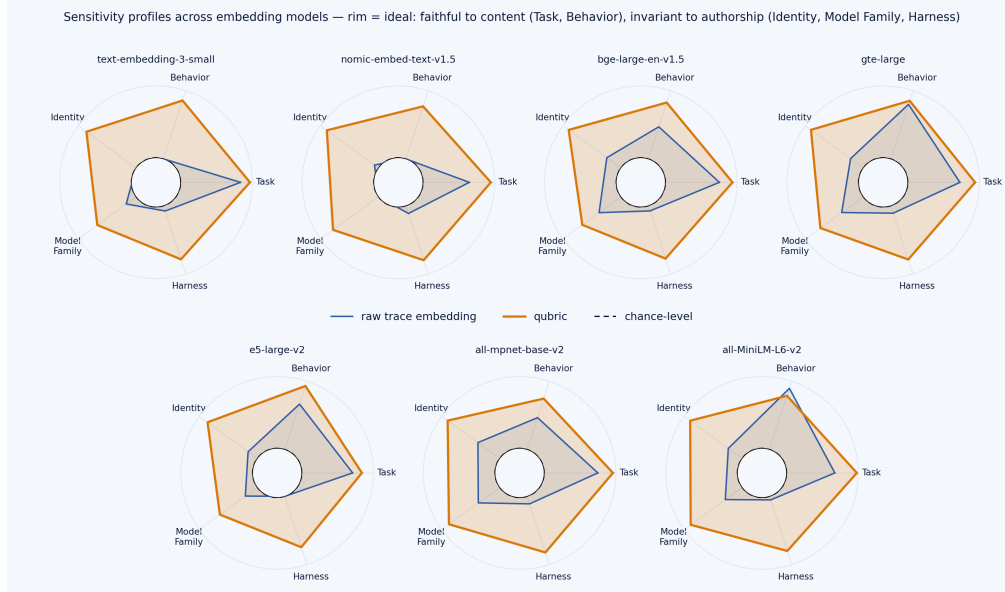


Figure 1: Sensitivity radars across seven embedders. Outside = desirable. Raw embeddings (navy) are authorship-dominated; qubric (amber) collapses Identity / Model family / Harness to chance while holding Task and Behavior.



Figure 2: QUENCH: error predicting true resolve rates vs. probe budget m (left) and vs. reference-pool size n (right); leave-one-LLM-out, 107 systems.